

RESEARCHER`S WEEKLY PROGRESS REPORTS

AMA COMPUTER COLLEGE-MALOLOS

ECE 201:DESIGN PROJECT 2 WEEKLY PROGRESS REPORT

PROJECT TITLE:	Automated Library Management System using Barcode and RFID
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Date Submitted	November 23, 2018
Period Covering	November 12, 2018 TO November 23, 2018
Over-all Project Status (%)	50%

Status	Milestone Description
Working	RFID and Barcode test
In Progress	Reading of the RFID UID on Visual Studio
Purchased	USB Barcode Scanner

Highlight Progress

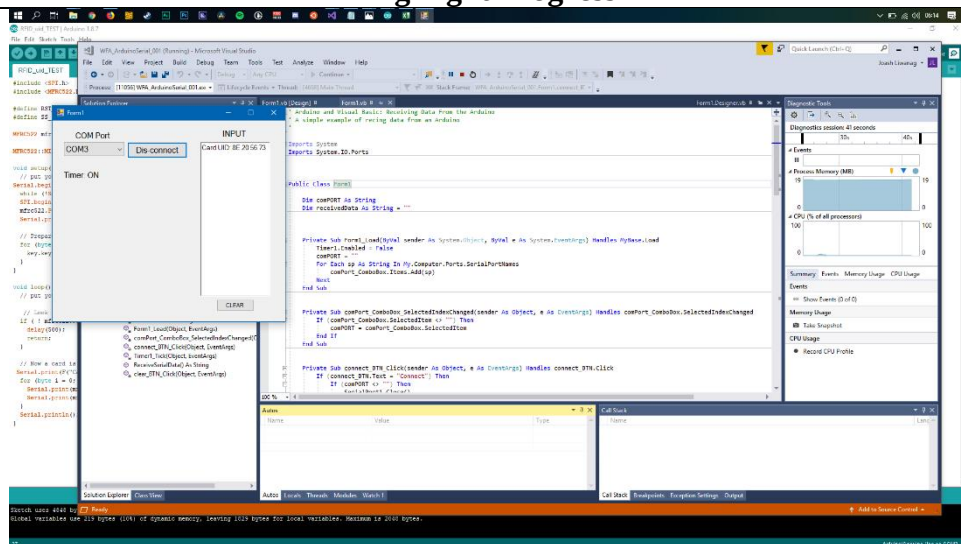


Figure 1. Reading of Serial output on a test Visual Basic program through COM Port 3.

```
RFID_uid_TEST | Arduino 1.8.7
File Edit Sketch Tools Help
Serial Monitor

RFID_uid_TEST

#include <SPI.h>
#include <MFRC522.h>

#define RST_PIN 9 // Configurable, see typical pin layout above
#define SS_PIN 10 // Configurable, see typical pin layout above

MFRC522 mfrc522(SS_PIN, RST_PIN); // Create MFRC522 instance

MFRC522::MIFARE_Key key;

void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600); // Initialize serial communications with the PC
  while (!Serial) { // Do nothing if no serial port is opened (added for Arduinos based on ATMEGA32U4)
    // Wait for serial port to be opened
  }
  SPI.begin(); // Init SPI bus
  mfrc522.PCD_Init(); // Init MFRC522 card
  Serial.println(F("Warning: this example overwrites the UID of your UID changeable card, use with care!"));

  // Prepare key - all keys are set to FFFFFFFFh at chip delivery from the factory.
  for (byte i = 0; i < 6; i++) {
    key.keyByte[i] = 0xFF;
  }
}

void loop() {
  // put your main code here, to run repeatedly:

  // Look for new cards, and select one if present
  if (!mfrc522.PICC_ReadCardSerial() || !mfrc522.PICC_ReadCardSerial()) {
    delay(500);
    return;
  }

  // Now a card is selected. The UID and SAM is in mfrc522.uid.
  Serial.print(F("Card UID:"));
  for (byte i = 0; i < mfrc522.uid.size; i++) {
    Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? "0" : "");
    Serial.print(mfrc522.uid.uidByte[i], HEX);
  }
  Serial.println();
}

Sketch uses 4548 bytes (15%) of program storage space. Maximum is 32256 bytes.
Global variables use 219 bytes (10%) of dynamic memory, leaving 1529 bytes for local variables. Maximum is 2048 bytes.
```

Figure 2. Arduino Code for reading the RFID UID.

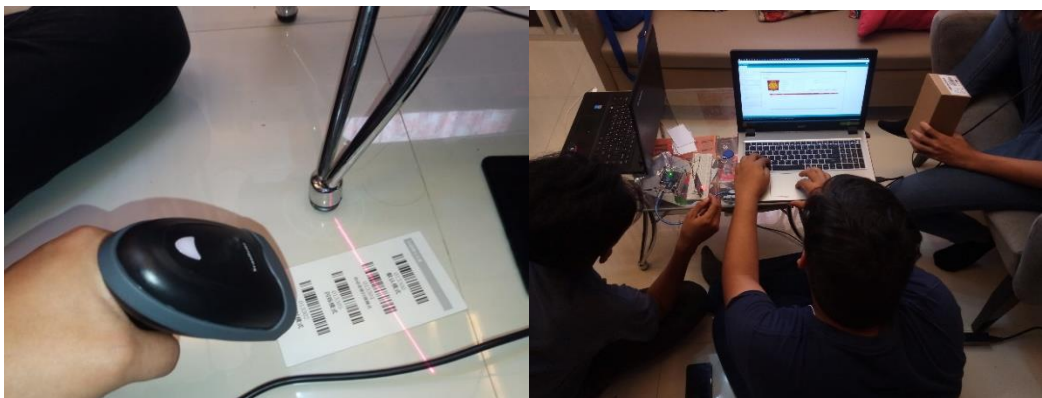


Figure 3. Testing out the Barcode Scanner